# BetterCode: Shareable

Publish codes that others can use

# Overview

- *Recap day 1*+*2*+*3*+4 - readable, reusable, reproducible, reliable
  - Help with being a better coder
  - Generate high-quality code
- **Shareable**: Github
  - Brag about your research
  - Activity 1: Publish Github repo
  - Activity 2: End-of-workshop quiz

# You **should** and **must** share your code & data (to publish)

- Transparency
- Collaboration & communication
- Requirements of many major journals

# CovidSim (Imperial College)

- Simulation model of COVID-19, influence major lockdown policies
- Released: March 2020
- Critism
  - Slow code release: April/May 2020
  - Messy code, legacy code
- Independent validation
  - Core conclusion holds
  - Workable
  - Largely reproducible

## INFLUENTIAL PANDEMIC SIMULATION VERIFIED BY CODE CHECKERS

**Model shown to be reproducible after software engineers called the underlying code 'a buggy mess'.**

**By Dalmeet Singh Chawla**

"Totally unreliable." "A buggy mess." Over the past month, software engineers have sharply criticized the code underpinning an influential coronavirus simulation by scientists at Imperial College London, one of several models that helped to sway UK politicians into declaring a lockdown. Some media articles even suggested that the simulation couldn't be repeated by others – casting further doubt on the study. Now, a computational neuroscientist has reported that he has independently rerun the simulation and reproduced its results.

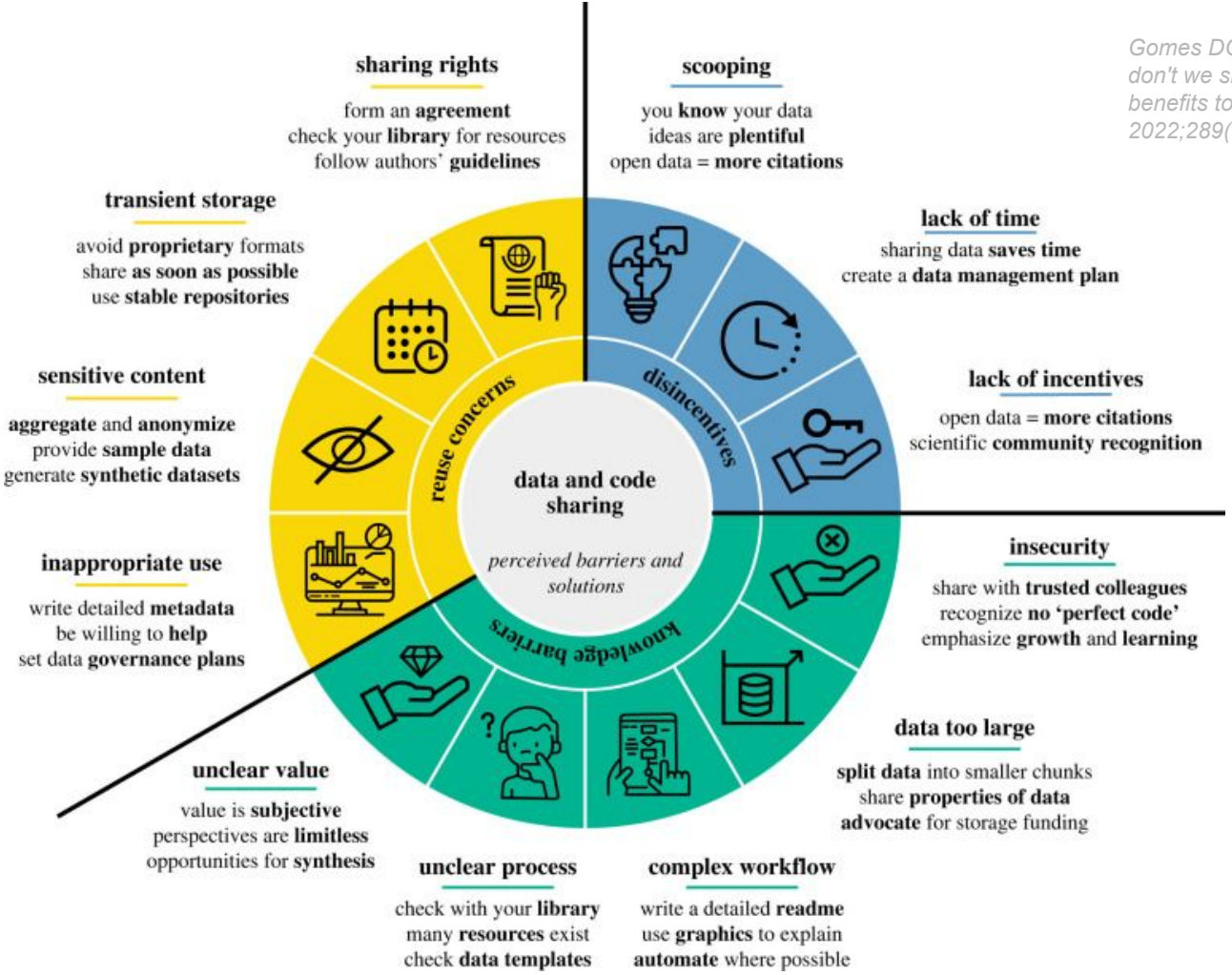
The successful code testing isn't a review of the scientific accuracy of the simulation, produced by a team led by mathematical epidemiologist Neil Ferguson. But it dispels some misapprehensions about the code, and shows that others can repeat the original findings.

The new test is "the best possible verification of Ferguson's simulations given the state of the art in computational science", says Konrad Hinsen, a computational biophysicist at the French national research agency CNRS in Paris, who was not involved in the work. In May, he wrote in a blogpost that the study's code looked "horrible", but that such shortcomings are expected in code written by scientists who aren't specialists in software development.

Released in mid-March, the original study reported that there could be half a million UK deaths if nothing were done to stop the virus, and modelled how policy interventions might help (N. Ferguson *et al.* 'Report 9' http://doi.org/ggqtdx; 2020). But Imperial scientists did not make the code available for public scrutiny. When a cleaned-up version was released

Nature | Vol 582 | 18 June 2020 | **323**

*Singh Chawla, Dalmeet. "Critiqued coronavirus simulation gets thumbs up from code-checking efforts." Nature vol. 582,7812 (2020): 323-324. doi:10.1038/d41586-020-01685-y*




*Gomes DGE, Pottier P, Crystal-Ornelas R, et al. Why don't we share data and code? Perceived barriers and benefits to public archiving practices. Proc Biol Sci. 2022;289(1987):20221113. doi:10.1098/rspb.2022.1113*

# Share work (code) that you can be proud of

**Internal Quality**

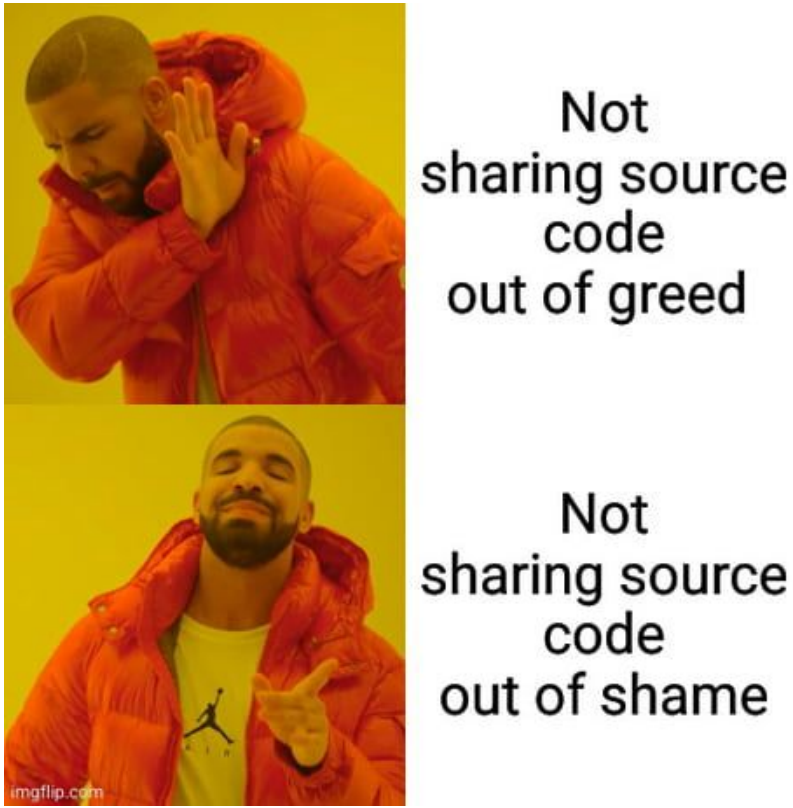
Readable

Reusable

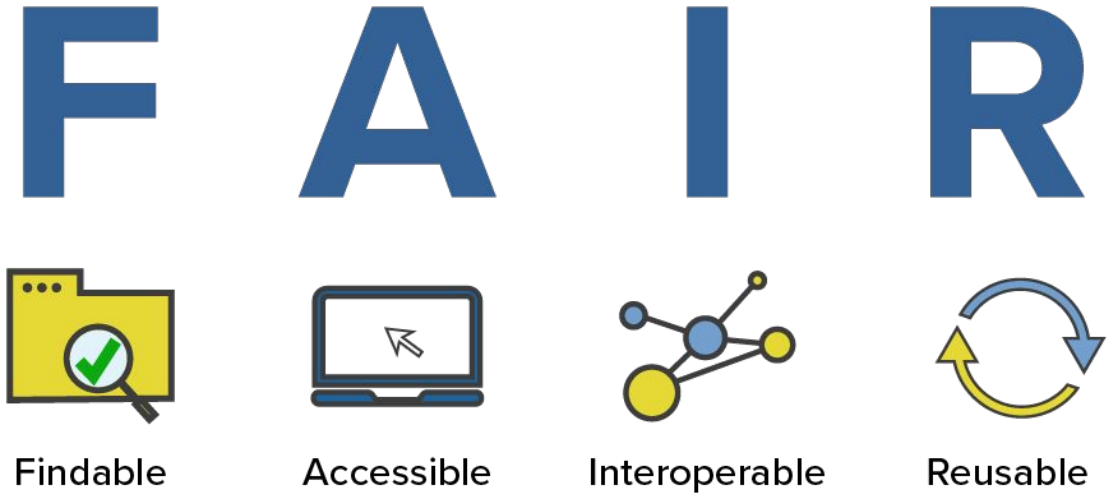
Reproducible

Reliable

**Standard Compliance?**



# Sharing Data: FAIR




2016: FAIR
2021: FAIR4RS (FAIR For Research Software)

# Sharing Code: TADA

Transferable
Available
Documented
Annotated

**Preprint**

*Ivimey-Cook ER, et al. (2025). TADA! Simple guidelines to improve code sharing. EcoEvoRxiv. DOI:10.32942/X2D93K*

MyCode.pdf

```
library(dplyr)
library(ggplot2)

data <-
read.csv("C:/mycomputer/caterpillar_data/data.csv")

summary_data <- data %>%
  group_by(habitat) %>%
  summarise(
    mean_count = mean(caterpillar_count),
    sd = sd(caterpillar_count),
  )

filtered_data <- data %>%
       filter(habitat != "D")

model1 <- glm(caterpillar_count ~ habitat,
  family = Poisson, data = filtered_data
)

figure1 <- ggplot(
  filtered_data,
  aes(x = habitat, y = catperillar_count)
) +
  geom_boxplot() +
  theme_bw()
```

Transferable
Available
Documented
Annotated

MyCode.R [T]

doi.org/... [A]

```
# Load packages#####
library(dplyr)
library(ggplot2)
library(here)

# Load caterpillar abundance data (w/o local file paths)#####
data <- read.csv(here("caterpillar_data", "data.csv"))

# summarise the mean number caterpillars with error#####
summary_data <- data %>%
  group_by(habitat) %>%
  summarise(
    mean_count = mean(caterpillar_count),
    sd = sd(caterpillar_count),
  )

# Remove values from habitat D as these are an error#####
filtered_data <- data %>%
       filter(habitat != "D")

# Run a Poisson general linear model######
# to analyse caterpillar abundance varying with habitat
# numeric results in "Caterpillar Abundance"
model1 <- glm(caterpillar_count ~ habitat,
  family = Poisson, data = filtered_data
)

#create figure 1, caterpillar count against habitat####
figure1 <- ggplot(
  filtered_data,
  aes(x = habitat, y = caterpillar_count)
) +
  geom_boxplot() +
  theme_bw()
```

[T]

[A]

README .txt [D]

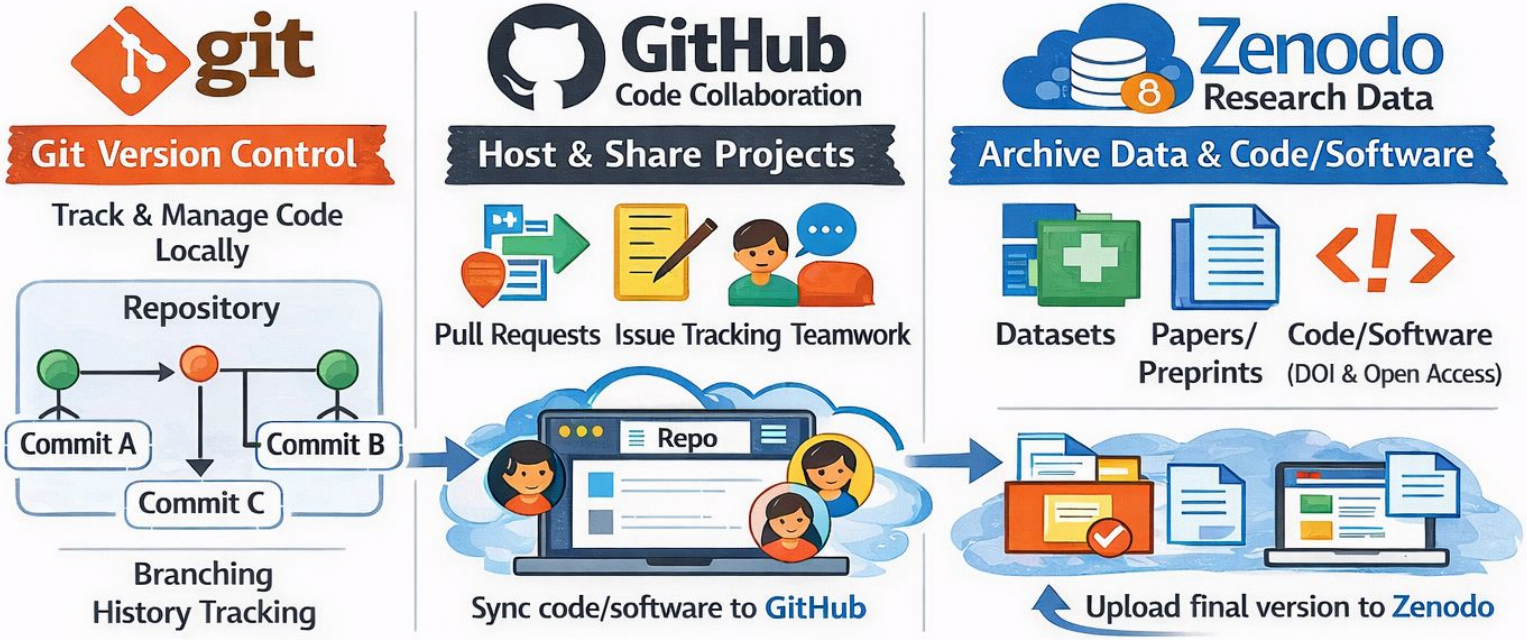
Authors: Ed Ivimey-Cook
Email: Ed@Ivimey-Cook.com
Title: Caterpillar abundance and habitat: A story.
Funders: SORTEE
Code License: MIT License in Code_License.txt

Code :
MyCode.R: Load packages, imports caterpillar_data, runs a poisson glm on filtered data. Produces Figure 1.

Software and Packages:
R v4.3.3
ggplot v2.3.2
dplyr v1.1.4
here v1.0.0

Data located here:
doi.org/...

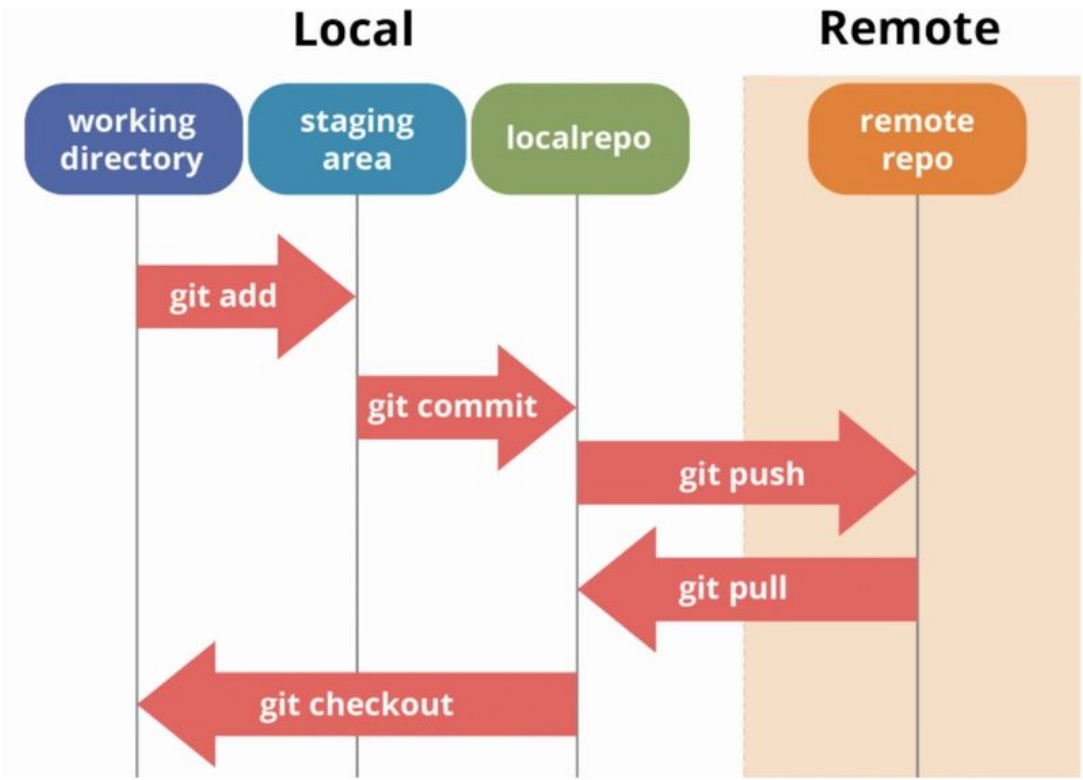
# Understanding Git, GitHub & Zenodo



# Demo 1: Local git - simple flow, no branch

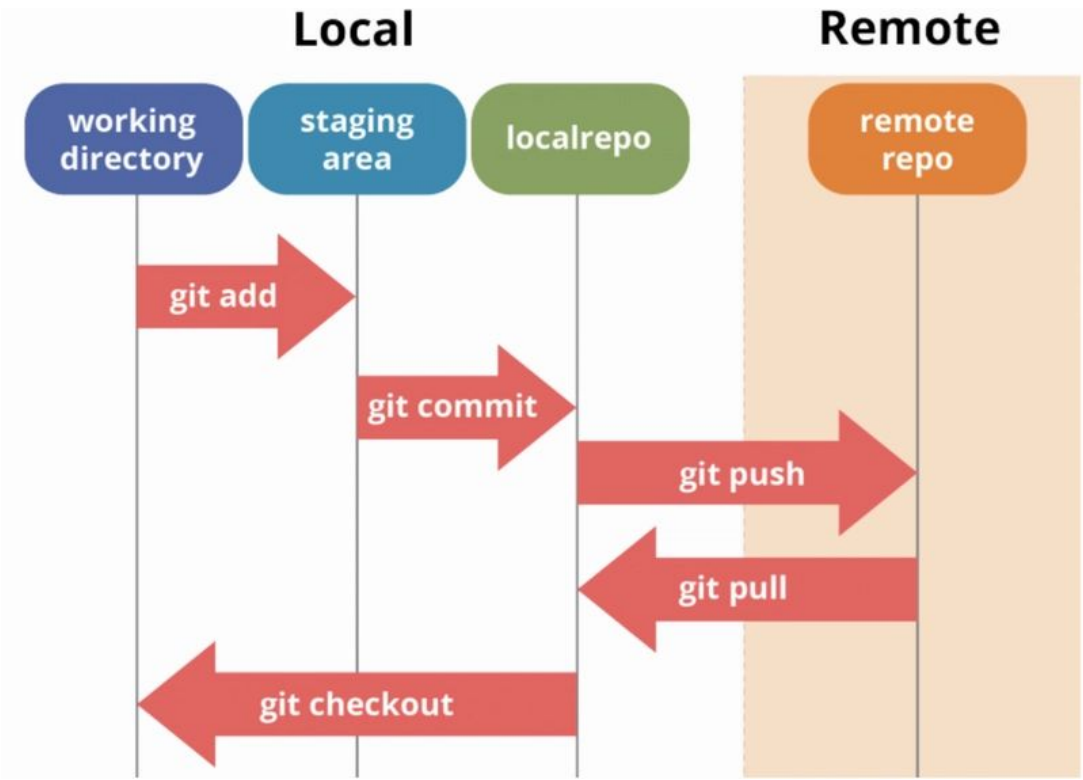
Check changes: git status
Check history: git log

The history is in the .git diretory



# Demo 1: Github - remote repositories

Git remote add origin



# Demo 1: Using IDE for version control with Git

# Demo 1: Github Repo

- Good organization
- README.md
  - What the project does
  - How to use it
  - Example usage
  - Basic structure
- .gitignore
  - Include scripts and environment description
  - Once pushed, it’s saved in the history

# Demo 1: Good Github Repo

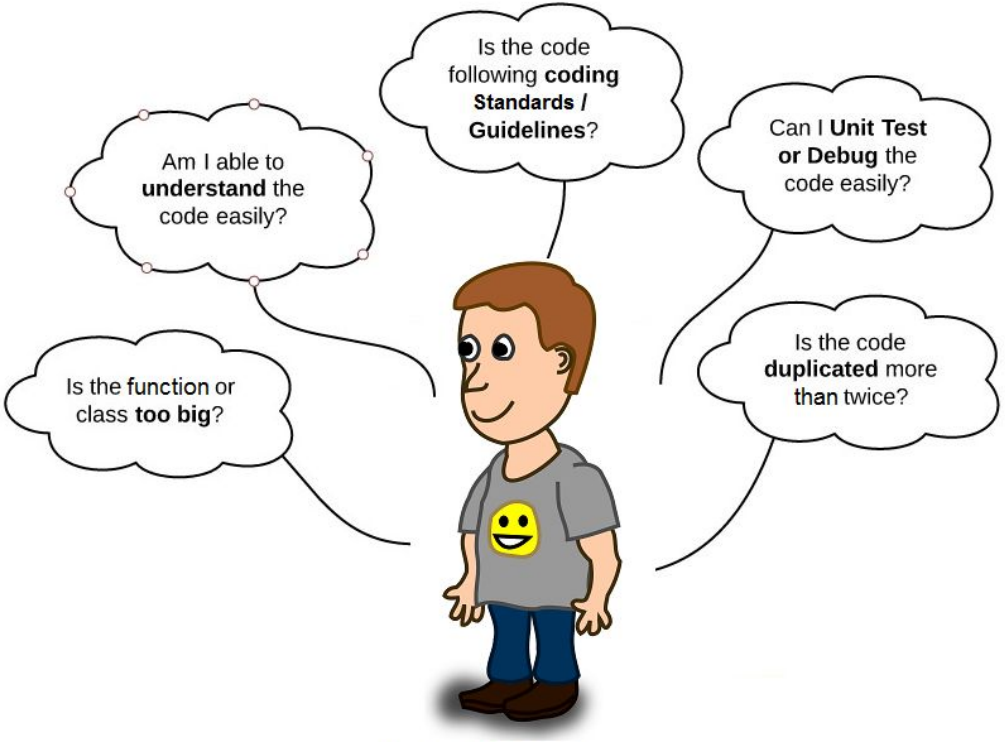
- Explain README.md, what should be included in
  - What the project does
  - How to use it
  - Example usage
  - Basic structure
  - Additional documents in /docs folder (if applicable)
- Explain .gitignore
- Explain versioning, note that deleting requires history clean up
  - Commit, push, pull
  - Do it within the IDE, on sumner2 too
- Show examples of real life Github repos: what is good, what is bad

# **Activity 1** - Publish your repo

- Create Github account
- Publish a directory
- Make changes
- Push, pull form it

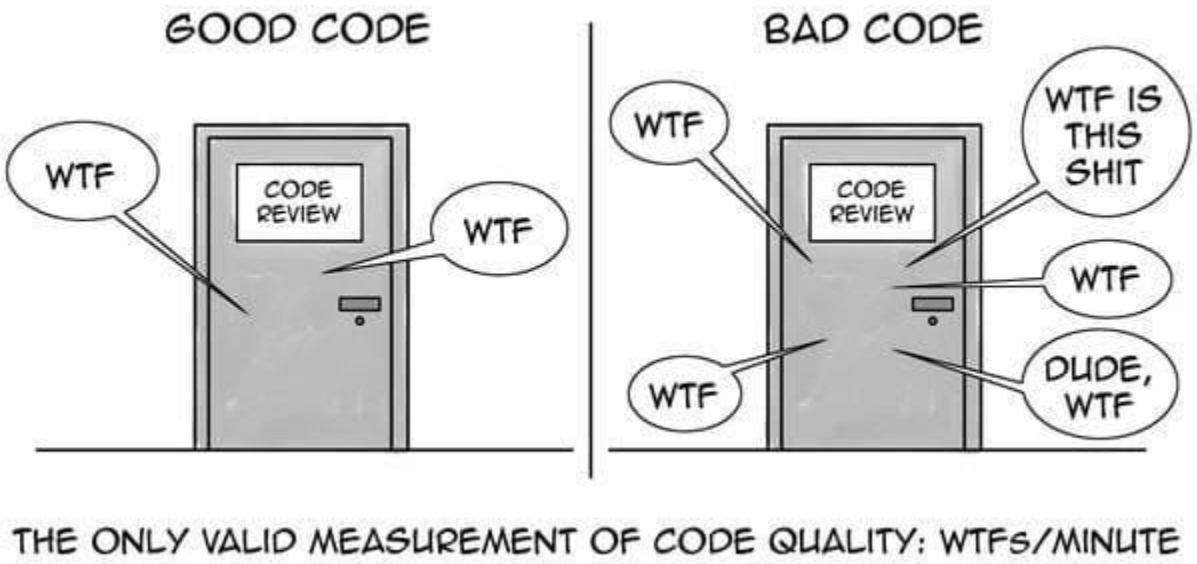
**Make it a habit, so make it as convenience as possible!!**

# Code Review




**Basic Code Reviewer**

# Demo 2 - Pull request



# Workshop Review

Questions

**Quizzes!!!**

# Food For Thought: Good Coding Practice in AI Era

Is coding & coding practice still important when AI writes most of the code?

# Take-away messages

Better code makes your research understood, trusted, and adopted by others

- Readability = clear communication
- Reusability = consistency and extensibility
- Reproducibility = verifiable results
- Reliability = trustworthy results
- Sharing = adoption and impact